

PROGRAMMABLE CONTROLLERS

MELSEC-F

FX Family Catalog



High-end Model

FX3U

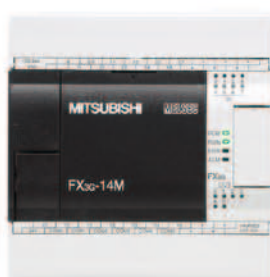


FX3UC



Standard Model

FX3GE NEW



FX3G



FX3GC



Entry level Model

FX3S NEW

FX3 series

Push the limits of control.

**Empowering
Industries**

FX3 series



Push the limits of control.

The Mitsubishi Electric FX PLC Family continues to be successful as a provider of customizable compact control solutions allowing customers to choose the best model to fit their applications.

Ease of use

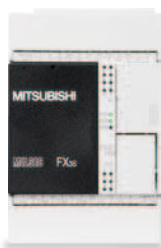
Control systems that require minimal setup and keep program development time short.

Affordable

A high performance to cost ratio makes economical design solutions for a diverse range of applications a reality. These features combined with Mitsubishi Electric's legacy in quality and reliability ensure that the 3rd generation of micro controllers will continue to be at the forefront of the compact PLC market and provide customers with a leading edge.

Models

Entry level Model



FX3S NEW

Simple and cost effective. Basic model that supports analog and communication expansion. Perfect for simple automation tasks.

Standard Model



FX3GE NEW



FX3G

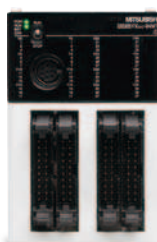
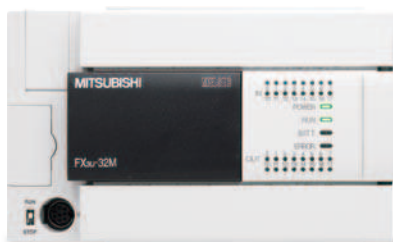


FX3GC

From automation to network, to more advanced control. Supports features required for basic control and a variety of applications.

Function Controllable I/O: 10 - 30 points

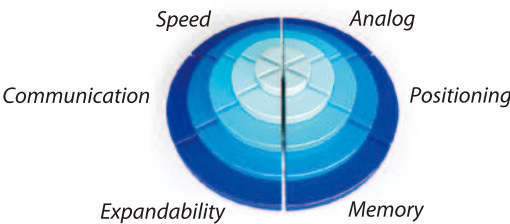
Controllable I/O: 14 - 128 points(Max. 256 with CC-Link remote I/O)





FX3U

Controllable I/O: 16 - 256 points
Max. 384 with CC-Link remote I/O
(Main Unit I/O: 16/32/48/64/80/128 points)



Top of the line

The FX3U is the original dual system-bus, high-speed, fully expandable compact PLC designed to seamlessly control communication, networking, analog, and positioning systems. With a maximum of 384 controllable local and networked I/O via CC-Link, the FX3U uses its power and flexibility to provide a solution for a variety of applications.

- 3rd generation compact PLC
- High efficiency with more speed, performance, memory, and new functions
- Built-in high speed processing and positioning
- The FX3U can control a maximum of 256 connected I/O, and up to 384 points with CC-Link remote I/O.

Product Details

All-in-one CPU, power supply and I/O. Includes many upgraded features from the FX2N, including high expandability using Expansion Boards and Special Adapters to add functionality.

Fast Instruction Times

Basic Instructions: 0.065 μ s / instruction (Contact Instruction)
Applied Instructions: 0.642 μ s / instruction (MOV Instruction)

Large Memory

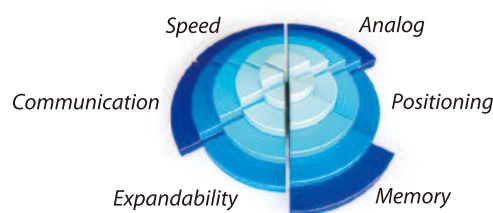
64,000 steps of built-in program memory.
Flash Memory Cassettes with loader function are available.

Applicable Standards

All products support EN and UL/cUL standards.
Various shipping approvals are supported as well.

Large Device Memory

Auxiliary Relays	7,680 points
Timers	512 points
Counters	235 points
Data Registers	8,000 points
Extension Registers	32,768 points
Extension File Registers	32,768 points
(with optional Memory Cassette)	



FX3UC

Controllable I/O: 16 - 256 points
Max. 384 with CC-Link remote I/O
(Main Unit I/O: 16/32/64/96 points)

Slim Fit

The FX3UC is the new ultra-compact high speed, fully expandable PLC. Based on 24V DC power and using connector-type transistor I/O, the FX3UC is designed for space conscious and modular applications. Created inline with the FX3U series, the FX3UC incorporates such attributes as built-in high speed I/O and the dual system-bus architecture, optimizing communication, networking, analog, and positioning control.

- 3rd generation super-compact PLC
- Reduced size and wiring using connector-type I/O
- Built-in high speed processing and positioning
- Even with its ultra-compact size, the FX3UC can be expanded to locally control up to 256 I/O, and up to 384 points with CC-Link remote I/O.

Product Details

Ultra-compact size main unit including many upgraded features from the FX2N, including high expandability using Special Adapters to add functionality.

Fast Instruction Times

Basic Instructions: 0.065 μ s / instruction (Contact Instruction)
Applied Instructions: 0.642 μ s / instruction (MOV Instruction)

Large Memory

64,000 steps of built-in program memory.
Flash Memory Cassettes with loader function are available.

Applicable Standards

All products support EN and UL/cUL standards.
Various shipping approvals are supported as well.

Large Device Memory

Auxiliary Relays	7,680 points
Timers	512 points
Counters	235 points
Data Registers	8,000 points
Extension Registers	32,768 points
Extension File Registers (with optional Memory Cassette)	32,768 points

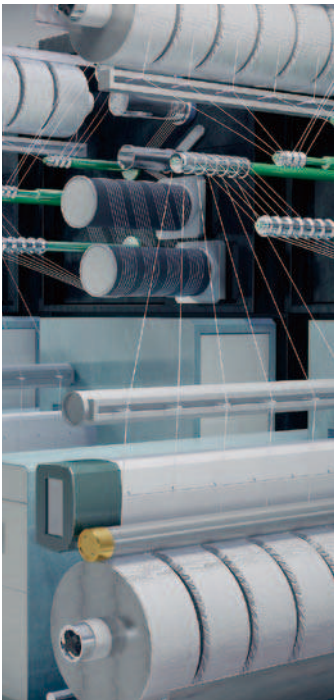
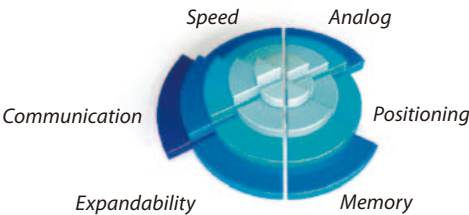


FX Main Units Lineup



FX3G

Controllable I/O: 14 - 128 points
Max. 256 with CC-Link remote I/O
(Main Unit I/O: 14/24/40/60 points)



Customized Control

The FX3G is an introductory compact PLC and is the newest addition to the FX3 series, designed for simple yet performance-critical applications. Incorporating innovative FX3 series technology the customer is presented with a suite of benefits.

- 3rd generation compact PLC
- Highly flexible
- Dual system-bus architecture
- Control of up to 128 directly connected I/O, or up to 256 I/O with CC-Link remote I/O.

Product Details

All in one CPU, power supply and I/O. Includes many upgraded features from the FX1N. Especially usage of the FX3 series ADP bus system and expansion boards (BD).

Instruction Times

Basic Instructions: 0.21 μ s / instruction (Contact Instruction)
Applied Instructions: 0.42 μ s / instruction (MOV Instruction)

Large Memory

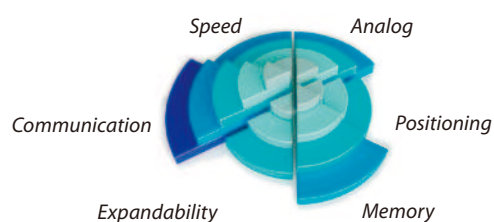
32,000 steps of built-in program memory.
EEPROM memory cassette with loader function is available.

Applicable Standards

All products support EN and UL/cUL standards.
Various shipping approvals are supported as well.

Large Device Memory

Auxiliary Relays	7,680 points
Timers	320 points
Counters	235 points
Data Registers	8,000 points
Extension Registers	24,000 points
Extension File Registers	24,000 points



FX3GC

Controllable I/O: 32 - 128 points
Max. 256 with CC-Link remote I/O
(Main Unit I/O: 32 points)

Super Compact Control

The FX3GC expands the FX3 series line up with a new connector type PLC entry model. Having the same processing capability as the highly successful FX3G, this super compact product reduces installation space, wiring time and potential wiring faults.

- 3rd generation super-compact PLC
- Reduced size and wiring using connector-type I/O
- Dual system-bus architecture
- Control of up to 128 directly connected I/O, or up to 256 I/O with CC-Link remote I/O.

Product Details

All in one CPU, power supply and I/O.

Same programming specification as the FX3G products in an ultra compact housing with connector type wiring for fast and fault free wiring.

Instruction Times

Basic Instructions: 0.21 μ s / instruction (Contact Instruction)

Applied Instructions: 0.42 μ s / instruction (MOV Instruction)

Large Memory

32,000 steps of built-in program memory.

Applicable Standards

All products support EN and UL/cUL standards.

Please consult with your local Mitsubishi representative regarding FX3GC shipping approvals.

Large Device Memory

Auxiliary Relays	7,680 points
Timers	320 points
Counters	235 points
Data Registers	8,000 points
Extension Registers	24,000 points
Extension File Registers	24,000 points



FX Main Units Lineup



Speed

Analog

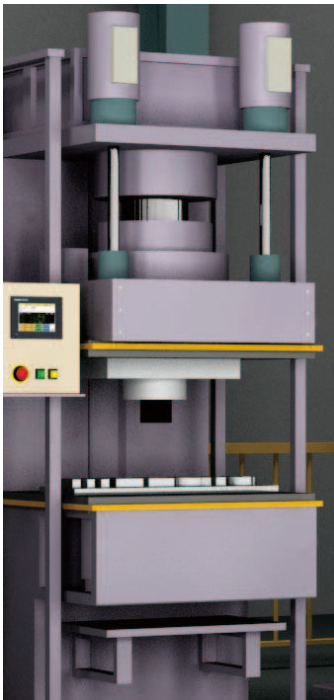
Positioning

Memory

Expandability

Communication

Controllable I/O: 24 - 128 points
Max. 256 with CC-Link remote I/O
(Main Unit I/O: 24/40 points)



All-in-one standard

FX3GE adds built-in analog input/output and Ethernet connectivity on top of FX3G performance. A great fit for many applications.

- 3rd generation, all-in-one PLC
- Highly flexible
- Dual system-bus architecture
- Control of up to 128 directly connected I/O, or up to 256 I/O with CC-Link remote I/O.

■ Product Details

All-in-one CPU, power supply and I/O, with integrated analog I/O and Ethernet port ready to use straight out of the box.

■ Instruction Times

Basic Instructions: 0.21 μ s / instruction (Contact Instruction)
Applied Instructions: 0.42 μ s / instruction (MOV Instruction)

■ Large Memory

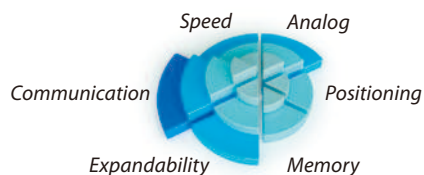
32,000 steps of built-in program memory.
EEPROM memory cassette with loader function is available.

■ Applicable Standards

All products support EN and UL/cUL standards.

■ Large Device Memory

Auxiliary Relays	7,680 points
Timers	320 points
Counters	235 points
Data Registers	8,000 points
Extension Registers	24,000 points
Extension File Registers	24,000 points



FX3S NEW

Controllable I/O: 10 - 30 points
(Main Unit I/O: 10/14/20/30 points)

New possibilities

The newly released FX3s adds extra expandability to the high cost performance of the venerable entry-level FX1s. FX3s makes it possible to utilize analog, Ethernet and MODBUS® functions even in small-scale systems.

- Basic controller for general applications
- High performance with minimal size

■ Product Details

All-in-one CPU, power supply and I/O. Includes many upgraded features from the FX1s. Especially usage of the FX3 series ADP bus system and expansion boards (BD).

■ Instruction Times

Basic Instructions: 0.21 μ s / instruction (Contact Instruction)
Applied Instructions: 0.5 μ s / instruction (MOV Instruction)

■ Memory Specifications

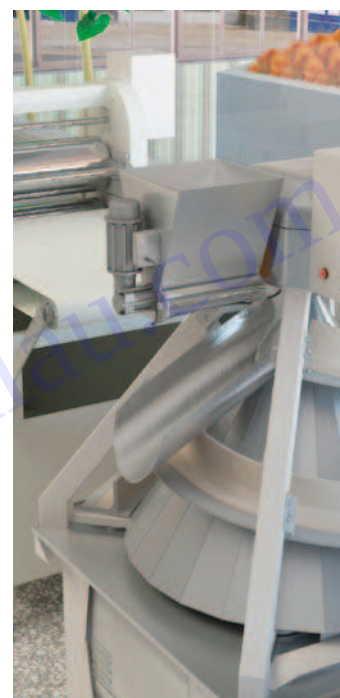
4,000 steps of built-in program memory.
No battery. No maintenance.

■ Applicable Standards

All products support EN and UL/cUL standards.
Please consult with your local Mitsubishi representative regarding FX3s shipping approvals.

■ Device Memory Specifications

Auxiliary Relays	1,536 points
Timers	138 points
Counters	67 points
Data Registers	3,000 points
Extension File Registers	2,000 points



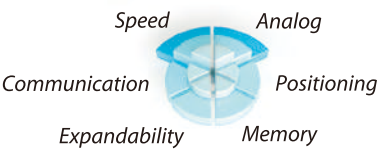
FX Main Units Lineup



FX1S

Controllable I/O: 10 - 30 points
(34 with FX1N-4EX-BD)

(Main Unit I/O: 10/14/20/30 points)



Keeping it simple

The FX1s is the fit-and-forget PLC solution for space and cost conscious applications requiring up to 30 I/O. This minimum-expandability, battery-less maintenance-free controller is perfect for handling straightforward processes and can be hidden away in locations without regular maintenance activities.

- Basic controller for general applications
- High performance with minimal size

■ Product Details

All-in-one CPU, power supply and I/O. Extension options includes a display unit and expansion boards for communication, analog control, or extra I/O.

■ Instruction Times

Basic Instructions: 0.7 μ s / instruction (Contact Instruction)
Applied Instructions: 3.7 μ s / instruction (MOV Instruction)

■ Memory Specifications

2,000 steps of built-in program memory.
No battery. No maintenance.

■ Applicable Standards

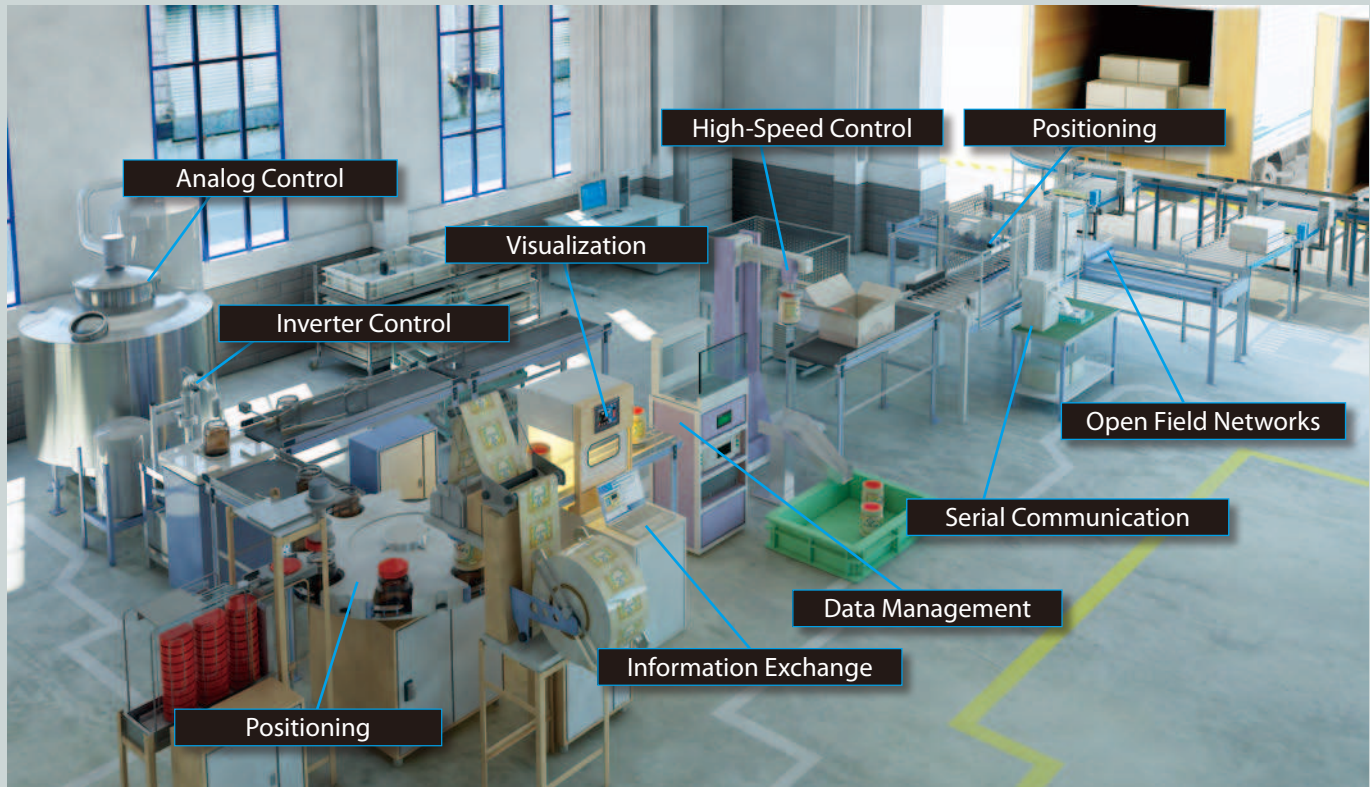
All products support various overseas standards.

■ Device Memory Specifications

Auxiliary Relays	512 points
Timers	64 points
Counters	32 points
Data Registers	256 points

FX Control Solutions

With superior built-in functionality and straightforward usability, the FX Series meets the needs of a variety of user applications. Using the factory image below as an example, key attributes that form the basis of automation control are demonstrated by the FX control system.



Analog Control



The FX Series features easy Analog to Digital and Digital to Analog conversion for all models using Expansion Boards, Special Adapters, or Special Function Blocks. For more information, go to Page 10.

High-Speed Control



Integrated High-Speed Counters

With 6 to 8 high speed counters on each model, the FX Series is perfectly suited for applications in need of pulse-catch functions, closed-loop feedback processing, or high-speed sensor use. For more information, go to Page 11.

Inverter Control



Integrated FREQROL Protocol

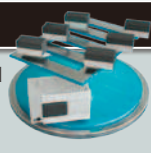
FX3 series PLCs contain the RS-485 communication protocol and special instructions that allow control of all Mitsubishi Electric FREQROL Inverters. For all other models, Analog inverter control can be used. For more information, go to Page 11.

Positioning



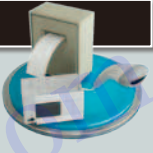
Built-in high-speed pulse outputs and special instructions enable the FX Series main units to control up to 3 independent axes of servo motion from the main unit itself. With special function blocks, interpolated and networked servo control solutions are also available. For more information, go to Page 12-13.

Open Field Networks



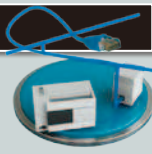
Among the networks supported by the FX Series are CC-Link and Ethernet, Modbus, and PROFIBUS, providing both a wide selection for new solutions and an interface into existing networks. For more information, go to Page 14.

Serial Communication



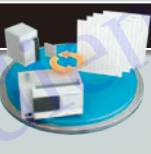
From PCs, printers, barcode readers, modems, and other PLCs, serial communication increases the flexibility of the FX Series system to allow reliable data exchange over a variety of connections. For more information, go to Page 15.

Information Exchange



Information can be sent to a higher level PC that constantly monitors production, raising the reliability and overall efficiency of the system. For more information, go to Page 16.

Data Management



A new special adapter, the FX3U-CF-ADP, for the FX3U and FX3UC enables data to be automatically written to a CF card at specified intervals or under certain conditions. The data is saved in universal CSV format with user-defined file names and automatic timestamps to enhance efficiency. For more information, go to Page 17.

Visualization



HMI options for the FX Series range from simple text-based Display Units to advanced graphical touch-screen displays, known as the GOT1000 Series Graphic Operation Terminals. GOT1000 models start at 3.7" and go up to 15". For more information, go to Page 18-19.

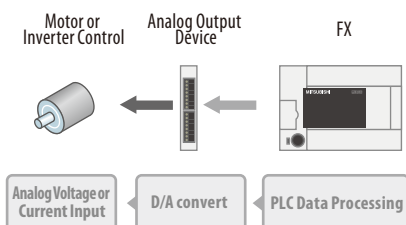
Analog Control

Expanding the FX PLC with analog inputs and outputs is required when the system needs to handle voltage or current inputs/outputs, temperature inputs, or temperature or PID control. The following expansion boards, special adapters, and special function blocks support different ranges and combinations of these features.

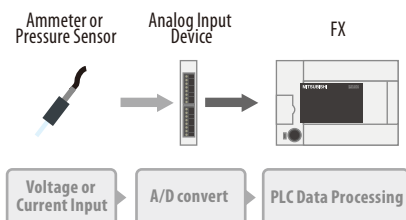


More details, please refer to the FX Analog Family.

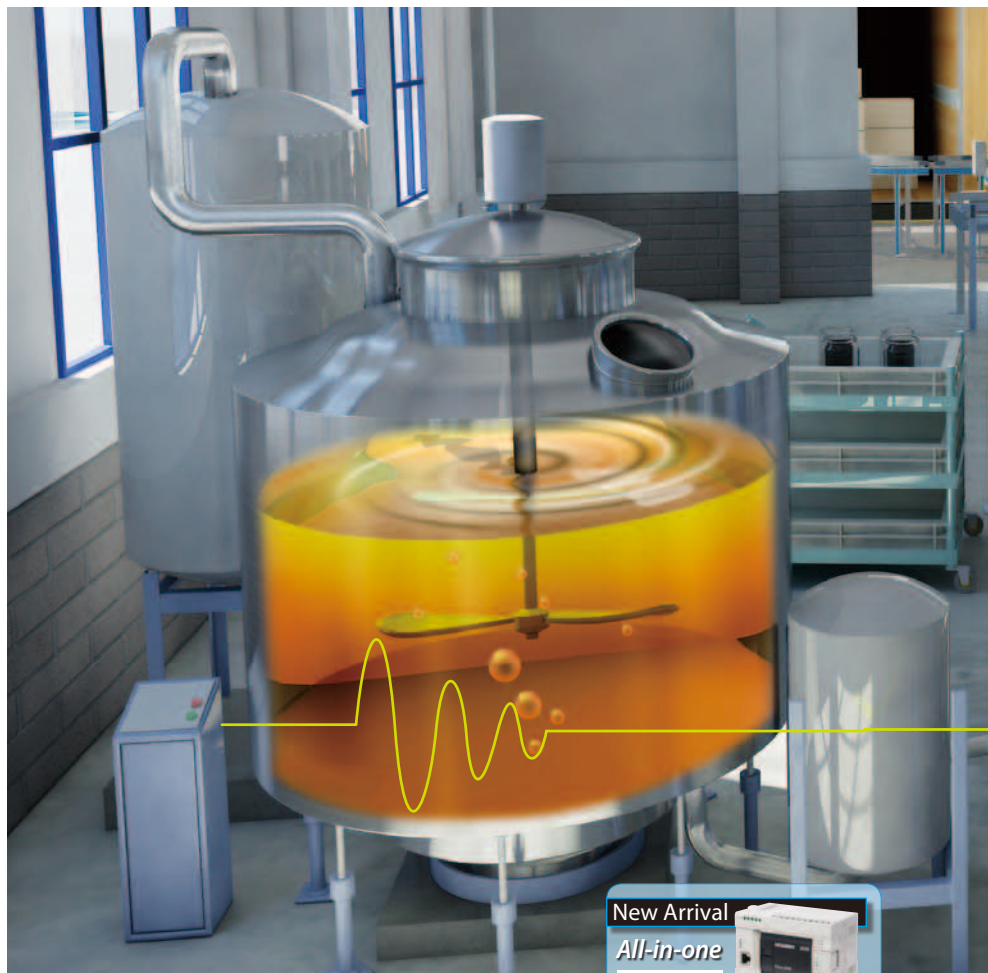
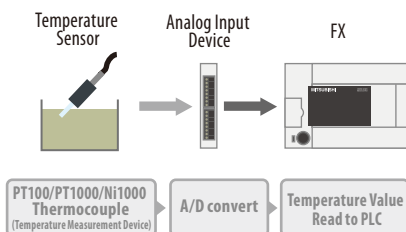
Analog Output



Analog Input



Temperature Input



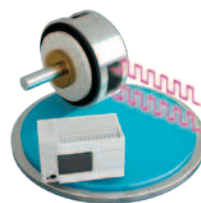
New Arrival
All-in-one
 Analog
 -Input 2ch
 -Output 1ch
FX3GE **NEW**

Analog Modules Lineup

	Expansion Boards	2ch	3ch	4ch	5ch	8ch
Analog Output	1ch FX1N-1DA-BD FX3G-1DA-BD	FX2N-2DA	1ch FX0N-3A FX3U-3A-ADP	FX2N-4DA FX3U-4DA FX2NC-4DA FX3U-4DA-ADP	1ch FX2N-5A	
Analog Input	2ch FX1N-2AD-BD FX3G-2AD-BD	FX2N-2AD	2ch FX2N-4AD FX3U-4AD	FX2NC-4AD FX3U-4AD-ADP FX3UC-4AD	4ch FX2N-8AD	
Temperature Input		FX2N-2LC	FX2N-4AD-TC FX2N-4AD-PT	FX3U-4LC FX3U-4AD-TC-ADP FX3U-4AD-PT-ADP FX3U-4AD-PTW-ADP FX3U-4AD-PNK-ADP		

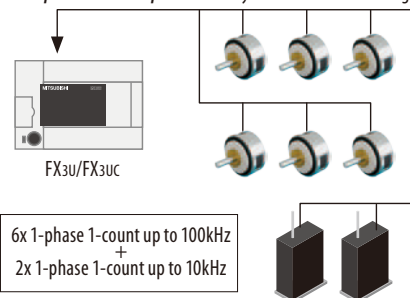
High-Speed Control

All FX Series PLCs are equipped with built-in open collector high-speed counters; up to 60 kHz for the FX3G/FX3GC/FX3GE/FX3S and up to 100 kHz for the FX3U/FX3UC. These high-speed inputs are connected to external devices such as encoders and ultrasound sensors for system feedback control. The FX3U main unit can be expanded with high-speed input special adapters, enabling differential line receiver inputs up to 200 kHz.



FX Integrated High-Speed Counter

Open Collector Inputs Rotary Encoder or Sensor Gauge

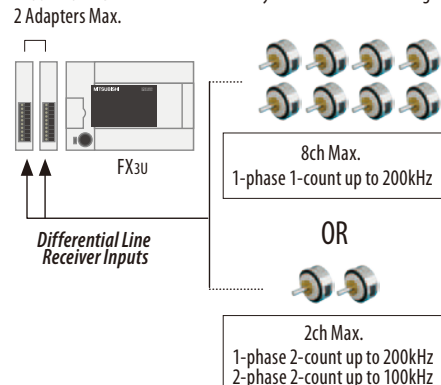


Note : Up to 50 kHz when using 2-phase 2-count

FX with High-Speed I/O Adapter

FX3U-4HSX-ADP
2 Adapters Max.

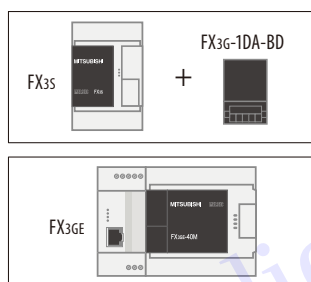
Rotary Encoder or Sensor Gauge



Note : When using high-speed I/O special adapters, the same numbered I/O terminals on the PLC main unit may not be used.

Inverter Control

Also known as Variable Frequency Drives (VFDs), Inverters play a vital role in many conveyor, pumping, and positioning systems. While the output frequency of the inverter can be varied easily using analog outputs on any of the FX Series PLCs, transfer of both parameters and commands between the PLC and Mitsubishi Inverters can be handled in FX3 series PLCs through serial communication using FREQROL protocol. Each RS-485 interface on an FX3 series PLC can control up to 8 Inverters with a total network extension of up to 500m when using the FX3U-485ADP-MB.

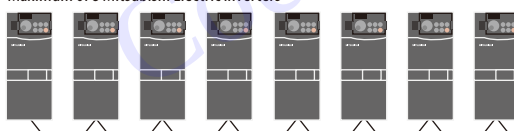


Analog Speed Control (0-10V DC, 4-20mA)

Inverter

Motor

Maximum of 8 Mitsubishi Electric inverters *



RS-485
Communication Equipment

FX3 series

RS-485
Communication

* : Applicable Inverters: F700/A700/E700/D700/V500/
F500/A500/E500/S500

Positioning

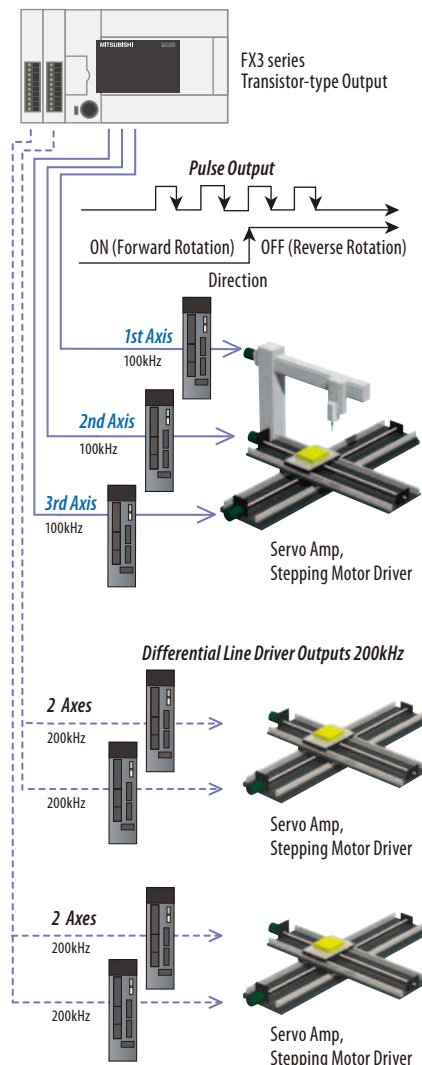
The built-in high-speed pulse outputs on all transistor-type FX Series PLCs, along with special positioning operations instructions, are designed to satisfy simple independent-axis positioning applications using servo and stepping motors with speed and precision. For more advanced applications like 4-axis control and noise-free interpolation, the FX system can be expanded with special adapters and special function blocks. With high performance at a low cost, the FX Series positioning possibilities provide a formidable package.



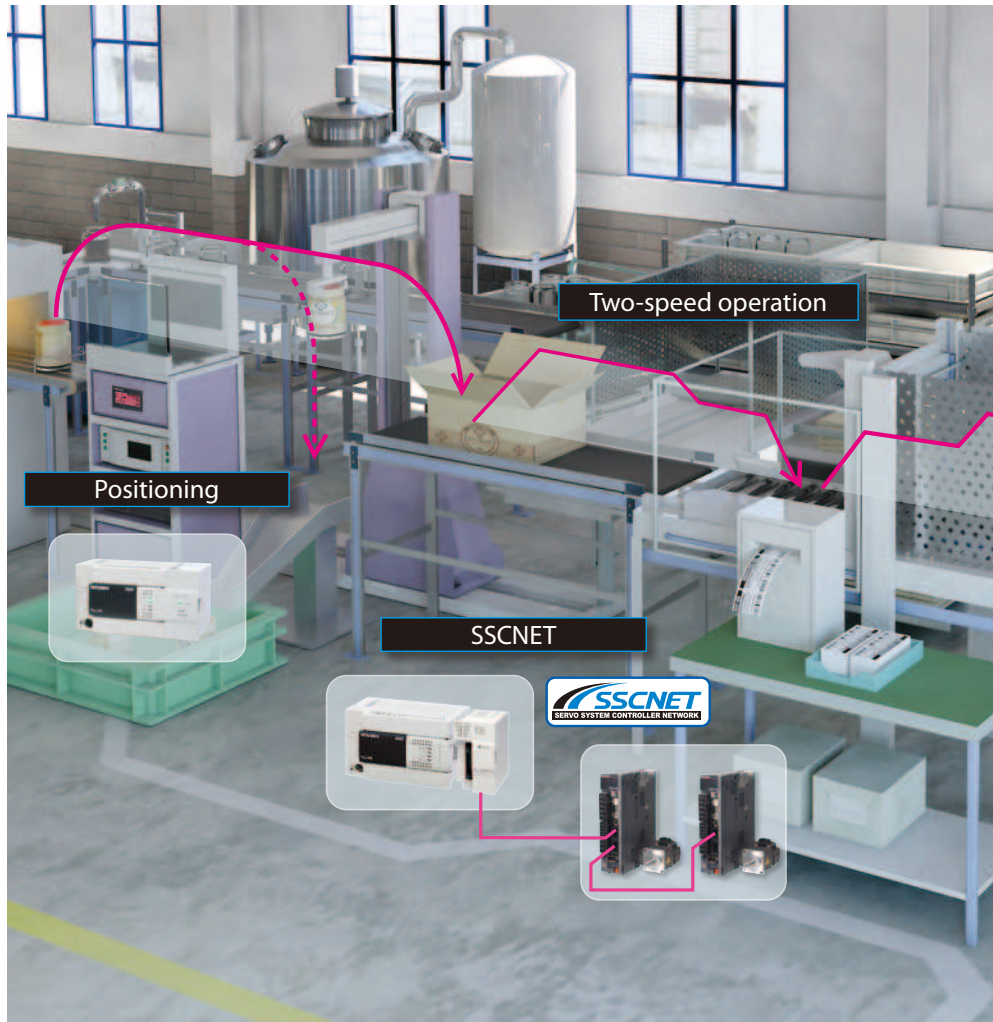
More details, please refer to the FX Positioning Family.

Upgraded Built-in Positioning Instructions for Easier Use

The FX Series allows for up to 3 axes of independent positioning control at up to 100 kHz. Connecting two high-speed output special adapters to the FX3U allows for up to 4 axes of control at up to 200 kHz. *

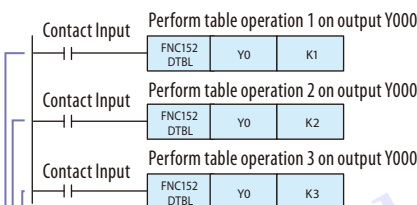


* : When using high-speed output special adapters, the same numbered I/O terminals on the PLC main unit may not be used.

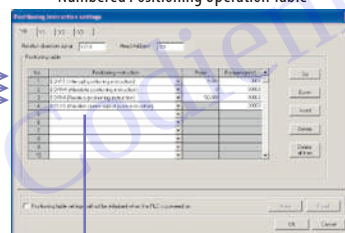


Easy Step-By-Step Positioning Programming Using the DTBL Instruction.

Only FX3U, FX3UC, FX3G, FX3GC, and FX3GE



Numbered Positioning Operation Table



Positioning Operation Setup

Positioning Modules Lineup



* : FX3U built-in axes are not available when the High-Speed Output ADPs are attached.